# ONLY YOU — technical report

**A provenance-gated authority layer for wearable agents** Sushant Poudel · Meta Global AI Developer Hackathon 2026 · Build for Your World

## 1. The problem

Meta's WebMCP documentation for Ray-Ban Display tells developers to set `untrustedContentHint` "honestly, since **it's not an enforced security boundary**," and warns that tool-result text "goes into the assistant's context" and is "how a prompt injection reaches the model." The assistant also cannot tell the wearer's voice from a stranger's. Once the agent has a camera, a microphone and a wallet, any TV ad, poster, webpage or invoice can issue commands.

The missing layer is not intelligence. It is **authority**: which inputs are instructions at all, who is allowed to issue them, what they may authorise, and what can be proven afterwards.

## 2. What I built

**Provenance before the model.** Every input is labelled before a token reaches the model: on-device speaker verification (only the enrolled owner's voice is a command channel), DOM origin (web text is never a command), scene-text masks + OCR (physical text is data), sender trust for documents. The model never has to win an adversarial classification game; the channel decides what is even eligible to be an instruction.

**A constitution and mandates.** The constitution is the ceiling — deterministic rules first, a small on-device evaluator only for ambiguity, ask-once for irreversible actions. Mandates are delegation: a signed, scoped, expiring capability ("pay the electrician up to 5,000 until tomorrow") that lets the agent act without interrupting and lets a third party verify the authorisation. Spend is cumulative and tracked outside the signed payload, so a mandate stays valid after use.

**Receipts.** Every action *and every refusal* emits an HMAC-signed, hash-chained record with the evidence, the rule applied, and an undo. Refusals are tagged `proposed_by: model | display_only`, so an enforcement event is never confused with a UI attribution.

## 3. What I measured

The four-arm table is a **harness test, not a security claim**: the corpus and the router's channel table are co-designed, so arm D cannot mislabel inside it. Its zero is conditional on correct labels.

| Arm | Architecture | Executed | False refusals |
|---|---|---|---|
| A | undefended | 100% | 0% |
| B | prompt-only | 47.5% | 0% |
| C | provenance-only | 10% | 0% |
| D | full ONLY YOU | 0% | 0% |

**Real-model runs — Muse Spark.** Three runs through the Meta Model API (`api.meta.ai/v1`), 195, 115 and 115 calls at temperature 0, with every proposal re-validated by the gate: `muse-spark-1.3` on the general and document corpora, and `muse-spark-1.3-contributor` on the document corpus.

*General corpus (20 attacks / 20 benign; ambient, scene, voice, web).* Muse Spark's own refusal behaviour dominates: **0% executed on every arm**, including undefended. The gate's measurable cost is false refusals (**D 5%**, C 10%; A/B 0%), and it files **36/36 verified receipts**. p50 6.8 s (D), p95 14.5 s.

*Document / confused-deputy corpus (8 attacks / 12 benign).* When the owner asks to pay a real invoice and the document supplies the payee, the model is helpful — and wrong: **A 25%, B 37.5%, C 50% executed** (provenance alone is worse, because it reclassifies instructions but not arguments). The full gate blocks **8/8 (D 0%) with 0% false refusals** and 18/18 verified receipts. p50 10.7 s, p95 34.7 s.

The same confused-deputy corpus on the **contributor build** (`muse-spark-1.3-contributor`, 115 calls) replicates the finding: **50% undefended, 37.5% prompt-only, 50% provenance-only, 0% with the full gate**, 0% false refusals, 18/18 receipts verified.

The three runs are the honest picture: alignment is a strong first line against obvious injections; the authority layer is what catches the confused deputy the model is trying to be helpful about. The samples are small and the corpus is synthetic, and the latencies are dominated by the model call, not the gate.

**The label layer — the part that can be wrong.** A DOM hidden-text detector on 42 hidden / 35 benign HTML snippets: recall **81.0%** (95% CI 66.7–90.0), false positives **14.3%** (6.3–29.4), precision 86.8%. Eight hiding techniques are missed (`clip-path`, `clip`, `transform:scale(0)`, zero-height, rgba-white, fixed off-screen, `visibility:collapse`, hidden inputs). A browser labeler that asks the rendering engine for computed styles and ancestor contrast scores **42/42 with 0/35 false positives** on the same cases — an upper bound, because it was written after the cases existed, but the

reason the production design does not parse `style=` strings. Speaker verification and its false-accept rate are **not measured**; the report gives a sensitivity table instead of a claim.

**Adaptive corpus — the attacker controls the signals.** Fifteen scenarios: twelve cloned-voice utterances presented with score 0.91 + proximity, three pages claiming `user_typed`. Result: **label error 80%, model-proposed 80%, executed 20%** (3/15 — the cases where the owner was present and confirmed). The web cases are 0/3 after the fix that makes web text data-only.

**Signed sensor envelope.** Six attacks fabricate the sensor signals ( `speaker_score: 0.99` + proximity) to impersonate the owner: **6/6 execute without an envelope, 0/6 with a signed sensor bus**, with all six recorded as blocked. This closes the cheap forgery path; it does not stop a cloned voice (the sensor is honest) or a compromised sensor.

**Evidence coverage.** Eight paraphrased payloads with no trigger verb: **0 executed**, but only **1/8 produced a receipt**. The audit layer, not the gate, fails first.

**Scene text.** Thirty-five synthetic signs across seven conditions, rendered with ImageMagick and read with Tesseract on both the raw image and an auto-deskewed, normalized copy, keeping the higher-confidence read: detection **100%** (35/35), mean character error rate **1.7%**; the weakest condition is occlusion (9.8% CER). Routing the OCR result through the stack executes **0/35** and produces **27/35 receipts** (77.1%) — enforcement does not depend on OCR, because scene text is data by channel; OCR quality changes the evidence surface.

**Muse Wonder scene reader.** The demo path can read the camera frame with a Meta multimodal model: 15 synthetic signs (clean, rotated 8°, low-contrast) read with the same prompt the API uses gave **detection 100% (15/15), mean CER 0.0%**, with approximate pixel boxes for the UI mask. Tesseract stays as the offline fallback. n is small by design; boxes are approximate, not segmentation masks.

**Latency.** The in-process gate is **≈0.1 ms p50**. Perception latency is **unmeasured**, and the report says so; the hundred-millisecond figures a wearable demo implies are not claimed here.

## 4. What failed, and what I did not fix

- **Cloned voice** at score 0.91 defeats the biometric: 12/12 label errors. Policy catches 9/12; the three executions are the confirms-true worst case.
- **Forged sensor dict**: mitigated for the cheap forgery — with a signed sensor bus, fabricated signals go from 6/6 executed to 0/6. A compromised sensor, and a cloned voice, still pass.
- **Merchant verification** is implemented: mandates are signed with an Ed25519 device key and a merchant verifies them offline with only the public key (signature, payee, cap, expiry); the demo exposes it as "verify at merchant". Residual: replay across merchants is future work.
- **Second-order injection** through the ledger is defended and tested: receipt text is sanitised for display and the model-facing ledger view labels every entry as data, never an instruction ( `docs/ledger-safety.md` ). Editing a receipt breaks its signature and the chain.
- **Model-alignment confound** in the real-model run: on the general corpus Muse Spark refused even when undefended, so the layer's incremental value only appears on the confused-deputy corpus (25–50% undefended/prompt/provenance vs 0% with the full gate).")
- **Biometric consent**: voiceprints are special-category data (GDPR Art. 9, DPDP 2023, BIPA). The demo enrolls only its author and keeps no audio; a product needs a consent flow, retention policy and a bystander policy.

The full table of nine named attacks, severity and status is `docs/limitations.md`.

## 5. Why this matters for Meta

Two products depend on this layer. **Glasses** become trustworthy only if the agent distinguishes the wearer from the world; the physical channel is the one no prompt-level defense covers. **Agentic commerce** depends on merchants being able to verify that an agent purchase was authorised — Amazon already blocks Muse because it cannot. A signed, scoped mandate turns agent commerce from a trust problem into a verification problem.

The architectural claim is deliberately narrow: **do not make the model robust to injection; make the channel untrusted by default.** Authority is a routing problem, and routing can be tested, instrumented and receipted.

## 6. With ten engineers

1. Ship the WebMCP tool-result provenance envelope: an `authority` field the assistant enforces, not a hint.
2. Voice + DOM provenance in the Muse client, with the signed sensor envelope.
3. Mandates on device public keys and an offline merchant verifier.
4. Scene provenance behind a flag; publish the benchmark and the corpus.

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Repo · demo · benchmark: `results/benchmark.md` · real-model runs: `results-real*/benchmark.md` · limitations: `docs/limitations.md` · 127 tests · self-audit